

Milestone 1: [TEAM UPDATE]

At this milestone, we have completed the cleaning, splitting, and visualization of the data. When importing, the template was already provided by HuggingFace. Our dataset needed a collate function to import the data and for it to be used for imaging and splitting into train/test/val. When splitting the data, we had to create a new augmentation for testing/val and training, due to the fact that different augmentation should be applied. When opening the images for inspection, we can see the quality of street-level imagery varying, as does the angle. They come from all types of different cameras, dashcams, phone cameras, or front doors. All of these can affect the accuracy of our predictions negatively. For that reason, we believe that augmentation to change the sky's location (Top, Left, Right) and some distortions could help the model the most.

We were able to set the figure size[20x20], apply a few transformations and augmentations (Resize, HFlip, RandomPerspective) and print our 100 images and their labels. For our set, we chose not to change the color in augmentations as that could have a negative effect since we are predicting based on the color of the sky, but the rest of the augmentations felt important to create a robust test of images. For our dataset, since we are pulling PM count from these images, we have to print the label element instead of just the label. In the future, we will be using this value to do a regression task to predict the class of AQI, so it feels especially important that the model is sensitive to color in images.

